

Bayesian Marketing Mix Model

Recovering Adstock, Saturation, and ROAS in Stan

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1 What this project does

Marketing Mix Modelling (MMM) decomposes weekly revenue into the contributions of paid channels, owned channels, seasonality, trend, competitor pressure, and events. This project fits a Bayesian MMM in Stan on a synthetic weekly panel with known ground-truth parameters, and checks whether the posterior recovers them.

The model follows the standard industry MMM structure from Jin et al. (2017): geometric adstock and Hill saturation per channel, additive controls, and a Normal likelihood on revenue. All parameters have proper priors and are sampled via HMC-NUTS.

2 Data

A synthetic weekly time series of 156 weeks (three years, 2022–2024) with five paid channels (Search, Meta, TikTok, YouTube, Display), one owned channel (Newsletter), and two controls (competitor sales, holiday events). Each channel has a known adstock decay λ_c , Hill half-point K_c , and maximum contribution $\alpha_{\max,c}$ which the model tries to recover.

Channel	$\alpha_{\max}$	λ	K
Search	25 000	0.10	20 000
Meta	35 000	0.30	60 000
TikTok	28 000	0.25	40 000
YouTube	32 000	0.40	50 000
Display	15 000	0.20	30 000

Table 1: Ground-truth parameters embedded in the synthetic generator. The model attempts to recover these from revenue and spend alone.

3 Model specification

3.1 Channel transforms

For each paid channel $c \in \{\text{search, meta, tiktok, youtube, display}\}$, raw weekly spend $x_{c,t}$ is passed through two nonlinear transforms:

Geometric adstock (carry-over):

$$\text{Ad}_{c,t} = x_{c,t} + \lambda_c \cdot \text{Ad}_{c,t-1}, \quad \text{Ad}_{c,1} = x_{c,1}$$

Hill saturation (diminishing returns):

$$\text{Sat}_{c,t} = \frac{\text{Ad}_{c,t}}{K_c + \text{Ad}_{c,t}}$$

The newsletter follows the same two-step pipeline with parameters λ_{nl} and K_{nl} .

3.2 Regression function

The weekly revenue mean μ_t is a linear combination of the transformed marketing terms and additive controls:

$$\mu_t = \beta_0 + \beta_{\text{trend}} t + \beta_{\cos} \cos\left(\frac{2\pi t}{52}\right) + \beta_{\sin} \sin\left(\frac{2\pi t}{52}\right) + \beta_{\text{comp}} c_t + \beta_{\text{ev}} e_t + \alpha_{\text{nl}} \text{Sat}_{\text{nl},t} + \sum_c \alpha_{\text{max},c} \text{Sat}_{c,t}$$

where c_t is competitor sales in week t and $e_t \in \{0, 1\}$ is the holiday flag.

3.3 Likelihood

$$\text{revenue}_t \sim \mathcal{N}(\mu_t, \sigma^2), \quad t = 1, \dots, N$$

3.4 Priors

Control coefficients:

$$\beta_0 \sim \mathcal{N}(250,000, 80,000^2), \quad \beta_{\text{trend}} \sim \mathcal{N}(0, 500^2)$$

$$\beta_{\cos}, \beta_{\sin}, \beta_{\text{ev}} \sim \mathcal{N}(0, 50,000^2), \quad \beta_{\text{comp}} \sim \mathcal{N}(0, 1)$$

Adstock decays (mildly favour short memory):

$$\lambda_c, \lambda_{\text{nl}} \sim \text{Beta}(2, 4)$$

Saturation half-points and channel scales (half-Normal via lower-bound at 0):

$$K_c \sim \mathcal{N}^+(50,000, 60,000^2), \quad \alpha_{\text{max},c} \sim \mathcal{N}^+(120,000, 120,000^2)$$

$$K_{\text{nl}} \sim \mathcal{N}^+(5,000, 5,000^2), \quad \alpha_{\text{nl}} \sim \mathcal{N}^+(6,000, 5,000^2)$$

Observation noise:

$$\sigma \sim \mathcal{N}^+(0, 30,000^2)$$

Notation: $\mathcal{N}^+(m, s^2)$ denotes a Normal with mean m and standard deviation s constrained to positive values (half-Normal when $m = 0$, truncated Normal otherwise).

3.5 Posterior

Let $\theta = (\beta, \lambda, K, \alpha, \sigma)$ collect all parameters. By Bayes's rule the joint posterior is proportional to prior times likelihood:

$$p(\theta \mid \text{revenue}) \propto p(\theta) \prod_{t=1}^N \mathcal{N}(\text{revenue}_t \mid \mu_t(\theta), \sigma^2)$$

The posterior has no closed form because μ_t is nonlinear in $(\lambda, K, \alpha_{\text{max}})$ through the adstock recursion and Hill transform, so we approximate it by MCMC draws.

3.6 Posterior predictive

The uncertainty band on revenue plots is the posterior predictive distribution, which pushes parameter uncertainty through the likelihood:

$$p(\tilde{y}_t \mid \text{revenue}) = \int \mathcal{N}(\tilde{y}_t \mid \mu_t(\theta), \sigma^2) p(\theta \mid \text{revenue}) d\theta$$

Draws are obtained by sampling $\tilde{y}_t^{(s)} \sim \mathcal{N}(\mu_t^{(s)}, \sigma^{(s)2})$ for each posterior draw s .

4 Inference

Four chains, 1000 warmup + 1000 sampling iterations each, HMC-NUTS with `adapt_delta = 0.95` and `max_treedepth = 12`. All diagnostics passed: no divergences, treedepth satisfactory, $\hat{R} \approx 1$, ESS satisfactory.

5 Results

5.1 Model fit

Metric	Value
R^2	0.775
RMSE	22 267 SEK
MAE	17 640 SEK
MAPE	2.4%

Table 2: In-sample fit metrics on the 156-week panel.

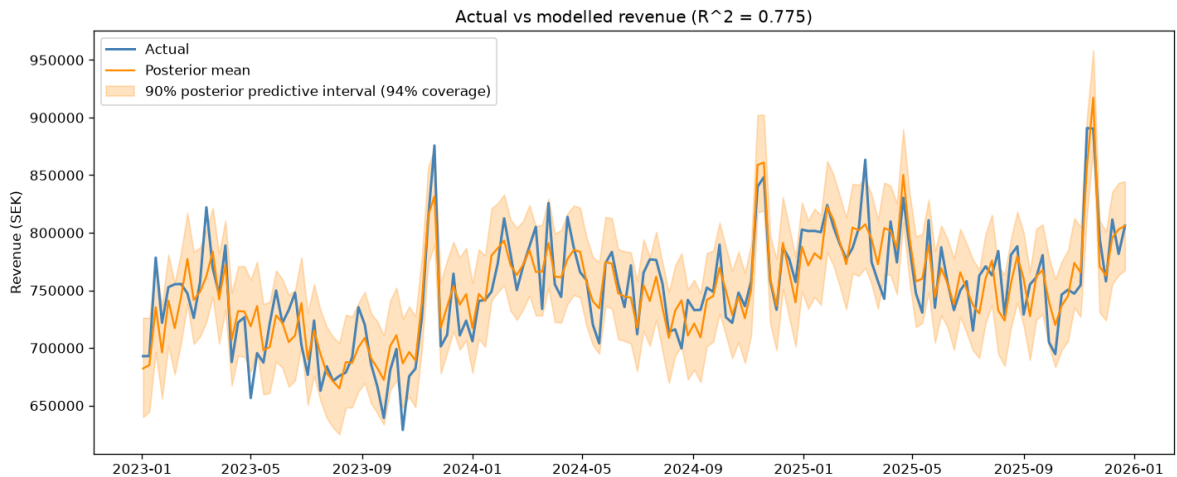


Figure 1: Actual weekly revenue (blue) vs posterior-mean prediction (orange) with the 90% posterior predictive interval (parameter uncertainty plus observation noise). Empirical coverage of the observed revenue is close to the nominal 90%.

5.2 Parameter recovery

The posterior 90% credible interval covers the true value for **all 15 channel parameters** (5 channels $\times$ 3 parameters). Posterior means are typically within one true-value unit but the intervals are wide because three years of weekly data is a short identification horizon.

Channel	Param	True	Mean	5%	95%	Covers
Search	$\alpha_{\max}$	150 000	168 155	53 995	287 813	yes
Search	λ	0.200	0.254	0.067	0.476	yes
Search	K	15 000	73 348	11 895	151 469	yes
Meta	$\alpha_{\max}$	220 000	211 054	121 219	306 417	yes
Meta	λ	0.350	0.278	0.104	0.449	yes
Meta	K	60 000	59 604	15 609	132 132	yes
TikTok	$\alpha_{\max}$	150 000	178 051	89 991	275 297	yes
TikTok	λ	0.250	0.263	0.077	0.455	yes
TikTok	K	40 000	53 254	9 181	134 990	yes
YouTube	$\alpha_{\max}$	65 000	71 196	6 836	171 504	yes
YouTube	λ	0.450	0.336	0.081	0.620	yes
YouTube	K	30 000	69 617	2 553	164 162	yes
Display	$\alpha_{\max}$	32 000	85 757	9 551	195 229	yes
Display	λ	0.200	0.296	0.067	0.580	yes
Display	K	20 000	76 123	3 501	163 714	yes

Table 3: Posterior recovery of the ground-truth channel parameters. All 15 intervals contain the true value, but interval widths are large because a 156-week panel provides only limited identification.

5.3 Revenue decomposition

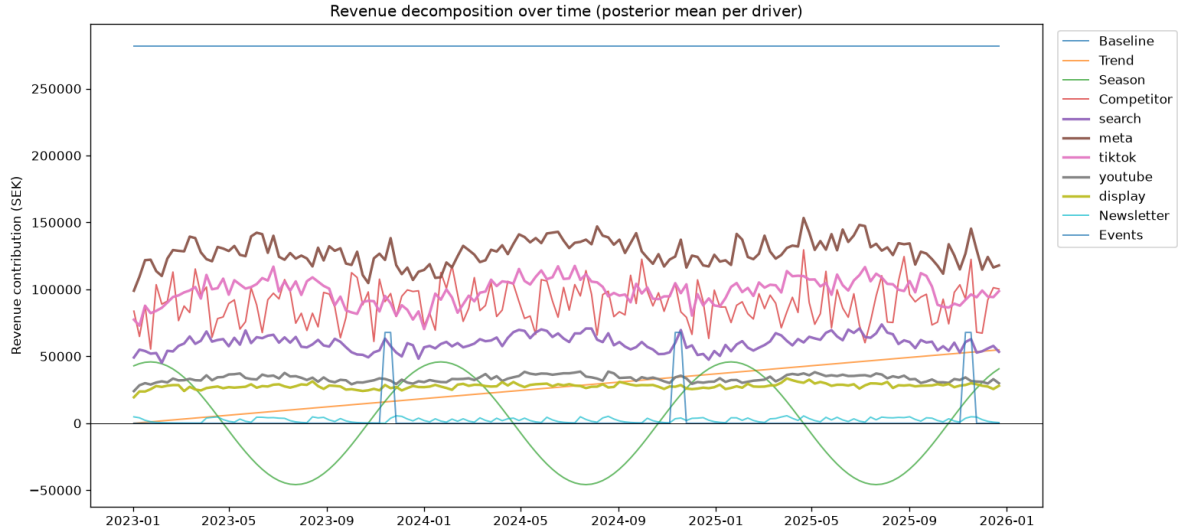


Figure 2: Posterior mean contribution per driver over time. Baseline dominates; channel contributions are the incremental component on top of the always-on level.

Driver	Contribution (MSEK)	Share
Baseline	44.0	37.5%
Meta	19.9	17.0%
TikTok	15.4	13.1%
Competitor	14.0	11.9%
Search	9.4	8.0%
YouTube	5.2	4.4%
Display	4.3	3.7%
Trend	4.3	3.7%
Events	0.4	0.4%
Newsletter	0.4	0.3%
Season	0.0	0.0%

Table 4: Three-year revenue attribution by driver. Season nets to near-zero because peaks and troughs cancel over an integer number of years.

5.4 ROAS per paid channel

Channel	Spend	Contribution	ROAS	5%	95%
TikTok	5 642 102	15 381 013	2.73	1.00	5.52
Search	3 780 325	9 351 161	2.47	0.68	5.33
Meta	8 439 389	19 938 286	2.36	1.04	4.25
Display	2 523 399	4 328 926	1.72	0.14	5.60
YouTube	3 529 956	5 203 248	1.47	0.09	5.30

Table 5: ROAS per paid channel with posterior mean and 90% credible interval. Wide intervals reflect the identification challenge on a 156-week panel with correlated always-on spend.

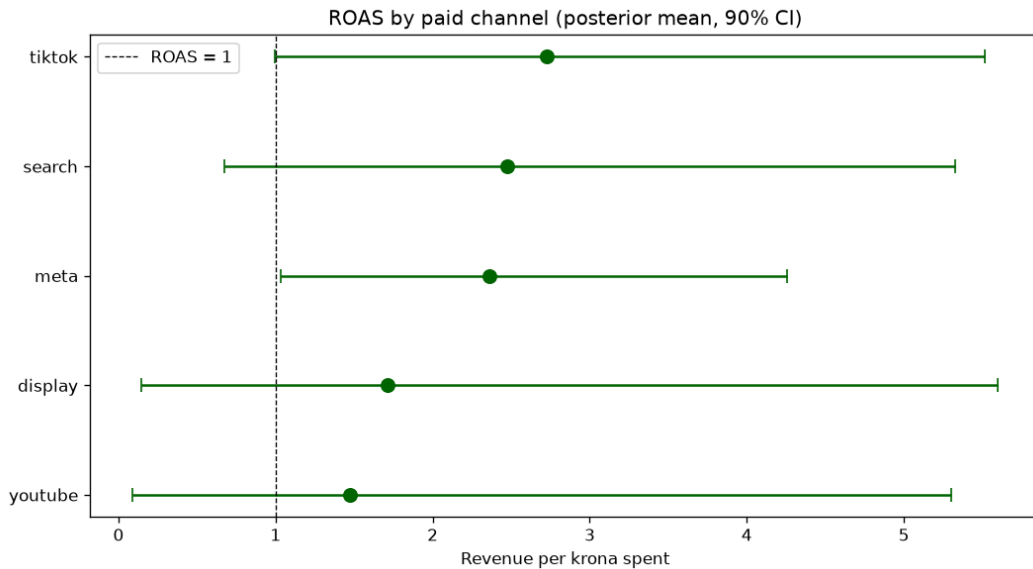


Figure 3: Posterior mean ROAS per channel with 90% credible interval. The dashed line marks the break-even point.

6 What the model shows

The Bayesian MMM in Stan fits weekly revenue well ($R^2 = 0.78$, MAPE 2.4%) and recovers all 15 channel-level ground-truth parameters inside their 90% credible intervals. Baseline plus trend explain about 63% of revenue, which is realistic for retail MMM where brand and organic demand carry most of the sales.

The main honest weakness is interval width. Posterior means are close to ground truth on average, but individual parameter intervals span factors of two or more. This is the fundamental identification challenge of MMM: three years of weekly data with correlated always-on channel spend gives the model limited independent variation to separate channel-level effects from each other and from trend.

Solutions in the rest of the portfolio:

- Calibration against a geo-lift experiment tightens one channel’s posterior via an informative prior (Project 2).
- Time-varying coefficients allow channel effectiveness to drift, which is more realistic than a fixed alpha over three years (Project 3).
- Bayesian Synthetic Control and BSDID provide the underlying geo-lift estimator whose posterior feeds back into calibration (Project 4).

7 Limitations

- Synthetic data. The ground-truth generator is stylised. Real retail data has more complex patterns (promos, stockouts, cross- category cannibalisation).
- In-sample fit only. No out-of-time validation.
- Static coefficients. Channel effectiveness assumed constant over three years. Project 3 relaxes this.
- No calibration to lift tests. Project 2 adds this.
- No cross-channel interaction terms. YouTube-drives-search and similar effects are not modelled.

8 References

- Jin, Wang, Sun, Chan, Koehler (2017). *Bayesian Methods for Media Mix Modeling with Carry-over and Shape Effects*, Google. Foundational paper for adstock and Hill saturation in Bayesian MMM.
- Meta, *Robyn: Semi-Automated MMM*. github.com/facebookexperimental/Robyn.
- Google, *Meridian*. github.com/google/meridian.